C1Q/SPP1 Macrophage Spatial Context as a Reproducible Traceability Feature in Non-Small Cell Lung Cancer: A Public Multi-Cohort Cancer-Informatics Evidence Synthesis

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Abstract

Objectives: Public cancer-informatics resources provide an opportunity to test whether macrophage-associated complement programs are reproducible markers of non-small cell lung cancer (NSCLC) tumor-microenvironment context. However, retrospective multi-omics analyses are vulnerable to overinterpretation when prognostic, spatial, and immunotherapy-response signals are presented as clinical tools or mechanistic proof.

Methods: We performed a conservative public-data evidence synthesis centered on the C1Q/SPP1 macrophage axis. Bulk transcriptomic and clinical data from TCGA NSCLC were used for discovery and model-behavior assessment after quality control. External prognostic traceability was evaluated in GSE31210 patients with complete follow-up, and small GEO immunotherapy datasets were analyzed separately as exploratory response-association cohorts. Spatial context was assessed using 10x Visium sections and Nanostring CosMx SMI data, with inference emphasized at the patient, section, or sample-contrast level rather than at the single-cell count level. Mendelian-randomization analyses were used for gene prioritization, not as definitive causal or therapeutic evidence.

Results: Nested cross-validation identified a stable 95-feature set enriched for immune and macrophage-context signals. C1QA, C1QB, and C1QC showed reproducible spatial autocorrelation in available spatial datasets (Moran's I approximately 0.35–0.39), whereas SPP1 showed stronger spatial clustering

(Moran's I = 0.712), consistent with a distinct macrophage-associated exclusion context. Retrospective model benchmarking showed statistically detectable but modest prognostic discrimination in TCGA and attenuated external performance in GSE31210; therefore, the model is reported as exploratory model behavior rather than as a validated clinical prediction tool. Small immunotherapy-treated cohorts showed directional response associations, but sample size and tumor-type heterogeneity preclude clinical claims.
Conclusions: C1Q/SPP1 captures a reproducible macrophage-spatial context in public NSCLC resources and may serve as a traceability feature for immune-active versus immune-excluded tumor-microenvironment states. The findings are hypothesis-generating and require prospective, spatially resolved, and experimental validation before clinical implementation or therapeutic targeting can be considered.

Keywords: Non-small cell lung cancer, Cancer informatics, C1Q, SPP1, macrophage context, spatial transcriptomics, reproducibility, traceability feature

1 Introduction

Non-small cell lung cancer (NSCLC) remains a major cause of cancer mortality, and immune checkpoint blockade has improved outcomes for a subset of patients [1–5]. However, PD-L1 expression, tumor mutational burden, and bulk immune-expression summaries do not fully capture the spatial and cellular organization of the tumor microenvironment (TME) [8, 9]. Spatial transcriptomics and single-cell technologies now make it possible to examine whether immune programs visible in bulk data also map to reproducible spatial contexts [14–16].

C1q, encoded by *C1QA*, *C1QB*, and *C1QC*, is produced largely by macrophage and dendritic-cell lineages in the TME. Prior work suggests that complement-associated macrophage states may accompany either immune activation or tumor-promoting inflammation, depending on context. *SPP1*-positive macrophage programs, by contrast, are often linked to stromal remodeling and immune exclusion. These observations motivate a narrower and more testable question: can the C1Q/SPP1 macrophage axis be traced reproducibly across public NSCLC transcriptomic, spatial, and clinical resources?

The present study is intentionally framed as a public cancer-informatics evidence synthesis. We do not claim to establish a clinical diagnostic assay, a treatment-selection tool, or a definitive mechanism. Instead, we evaluate whether C1Q/SPP1 behaves as a reproducible traceability feature linking macrophage context, spatial organization, retrospective survival associations, and exploratory immunotherapy-response signals. This conservative framing separates data-supported observations from hypotheses that require prospective and experimental validation.

2 Methods

2.1 Study Design and Evidence Hierarchy

This study used a staged, retrospective public-data design [25–27]. The evidence hierarchy was defined before interpretation: (1) TCGA NSCLC was used for discovery and internal model-behavior assessment; (2) GEO cohorts were used for external prognostic traceability where complete follow-up was available; (3) Visium and CosMx SMI datasets were used to evaluate spatial context, with inference emphasized at the patient, section, or paired-contrast level; and (4) immunotherapy-treated cohorts were treated as exploratory response-association datasets because of small sample sizes and tumor-type heterogeneity (Figure 1).

2.1.1 Data Acquisition and Quality Control

Discovery Cohort (TCGA): TCGA LUAD and LUSC transcriptomic and clinical data were retrieved from the NCI Genomic Data Commons [28, 29]. A total of 1,100 TCGA NSCLC tumor samples were initially considered (595 LUAD and 505 LUSC). After quality control and availability filtering, 1,038 samples contributed to discovery analyses. Analyses requiring complete survival, stage, or model features used the corresponding complete-case subsets; denominators are reported with each analysis.

Spatial Transcriptomics Data: For spatially resolved context analysis, we incorporated 10x Genomics Visium and Nanostring CosMx SMI datasets. Quality control, normalization, and platform-specific processing are described in the Supplementary Methods and source-data manifest.

- **10x Visium:** Included 8 tissue sections from 1 patient (Supplementary Table S12), utilizing a 55 μm spot diameter for tissue-level spatial mapping.
- **CosMx SMI:** Encompassed 8 tissue sections from 5 NSCLC patients, containing approximately 800,000 profiled cells targeted with a 960-gene RNA panel. Cell counts were treated as descriptive observations; independent inference was anchored to samples, sections, and paired contrasts.

External Cohorts: Clinical and expression data were retrieved from GEO. GSE31210 contained 226 lung adenocarcinoma samples after exclusion of non-tumor controls and 178 patients with complete follow-up for survival analysis. GSE91061, GSE126044, and GSE135222 were used only for exploratory immunotherapy-response association; GSE91061 was interpreted as cross-tumor support because it is not an NSCLC-only cohort.

2.2 C1Q/SPP1 Traceability Framework

We implemented a multi-stage traceability framework to evaluate whether C1Q/SPP1 macrophage context is reproducible across data layers:

Phase 1: MR-based gene prioritization: Mendelian randomization (MR) was used to prioritize genes with genetically supported associations [33], not to prove mechanism or treatment relevance. We used eQTL data from eQTLGen [35] and GTEx v8 with TwoSampleMR (v0.5.6), and inspected sensitivity diagnostics before assigning confidence.

Phase 2: Retrospective model-behavior assessment: Machine-learning algorithms were benchmarked to assess whether the stable feature set carried survival-associated signal. The resulting models were not interpreted as clinical tools and no fixed clinical cutoff is proposed.

Phase 3: Spatial context characterization: A C1Q/SPP1-centered scoring approach was used to evaluate macrophage-associated spatial organization and CD8-marker context in Visium and CosMx data.

2.3 Neighborhood Enrichment Analysis

Neighborhood enrichment analysis was performed in CosMx SMI data to describe local co-occurrence between C1Q-associated macrophage states and CD8-marker-positive contexts. A 50 μm radius was used around each index cell, and observed neighborhood composition was compared with a permutation-based null distribution (N=1,000 permutations). Because the number of independent patients was limited, these results were interpreted as local spatial-context support rather than definitive evidence of cell-cell signaling or causal recruitment.

2.4 Data Preprocessing and Statistical Analysis

2.4.1 Transcriptome Data Preprocessing

Bulk transcriptomic datasets were processed using dataset-appropriate normalization. TCGA RNA-seq expression values were filtered for low expression, log-transformed where appropriate, and aligned by gene symbol. External microarray data were quantile normalized or used as normalized series-matrix data when provided by GEO. Cross-platform analyses used within-cohort z-score scaling after feature alignment; platform-specific effects were not interpreted biologically. Excluded samples and feature-retention decisions are documented in the source-data manifest.

2.4.2 Statistical Analysis

All statistical analyses were performed using R and Python. Python-based data manipulation and visualization used Pandas [37], NumPy [38], SciPy [39], Matplotlib [40], and Seaborn [41]. Machine-learning tasks used Scikit-learn [42]. Survival analysis used Kaplan–Meier curves and Cox proportional hazards models. Model performance was evaluated with time-dependent AUC, concordance index (C-index), and hazard ratios with 95% confidence intervals where available. Data and code availability are detailed in the Declarations section.

2.5 Sample Size and Statistical Power

The TCGA discovery cohort (N=1,038 after quality control) was adequate for exploratory feature discovery and internal model-behavior assessment. External cohorts were smaller and had lower event counts; therefore, external survival and immunotherapy-response results were interpreted as traceability or exploratory association rather than confirmatory validation. No prospective sample-size calculation was possible because the study used public retrospective data.

2.6 Nested Cross-Validation Framework

To reduce data leakage and estimate feature stability, we implemented a 5×5 nested cross-validation framework.

Outer Loop (Model Evaluation):

- 5-fold stratified cross-validation (StratifiedKFold)
- Stratification basis: Vital status (alive/dead)
- Each fold: Training set 880 cases (80%), Test set 220 cases (20%)
- Random seed: 42 (ensuring reproducibility)

Inner Loop (Feature Selection):

- 5-fold cross-validation within each outer training set
- LASSO-based feature selection with α optimization using 5-fold cross-validation and 10-fold grid search (α range: 10^{-5} to 10^{-1}, step size: 0.01). Optimal alpha was selected based on minimum cross-validation error, with final $\lambda = 0.002$ yielding 91 non-zero coefficients (Supplementary Table S2).
- Features selected in 100% of outer folds (frequency = 5/5) retained as "core features"

This double-layer strategy separated feature selection from held-out fold assessment within TCGA. It does not replace independent prospective validation.

2.7 Mendelian Randomization for Gene Prioritization

We used two-sample Mendelian randomization [44] (MR), implemented in the *TwoSampleMR* R package, as a gene-prioritization layer. MR estimates were interpreted as genetically supported associations under instrumental-variable assumptions, not as proof of spatial mechanism, therapeutic actionability, or clinical utility.

Instrumental Variable Selection

- Genome-wide significance threshold: $P < 5 \times 10^{-8}$
- Linkage disequilibrium (LD) pruning: $r^2 < 0.001$, window size = 10,000 kb
- F-statistics calculated for all IVs, with $F > 10$ indicating strong instruments

MR Methods

- **Inverse Variance Weighted (IVW)**: Primary summary estimate under standard MR assumptions.
- **MR-Egger**: Used to assess directional pleiotropy and provide a sensitivity estimate under weaker assumptions.
- **Weighted Median**: Used as a sensitivity estimate when a subset of instruments may be invalid.

Comprehensive Sensitivity Analyses:

1. **Instrument strength**: F-statistics (median $F = 52,096.3$, 100% strong IVs with $F > 10$)
2. **Horizontal pleiotropy**: MR-Egger intercept test (18.5% of genes showed no pleiotropy, $P > 0.050$)
3. **Heterogeneity**: Cochran's Q test (62.3% showed no significant heterogeneity, $P > 0.050$)
4. **Outlier detection**: MR-PRESSO identified and removed outlier variants (70.0% of genes had no outliers, $P > 0.050$)
5. **Leave-one-out analysis**: Iteratively excluded each SNP to assess single-variant influence.

Exposure and Outcome Definitions: In all MR analyses, the exposure was defined as eQTL-mediated gene expression levels in immune-specific contexts, and the outcome was defined as the proportion of immune cell infiltration (e.g., $CD8^{+}$ T-cell density) or specific immune phenotypes in lung cancer tissues.

Detailed MR diagnostic parameters and leave-one-out sensitivity plots are provided in Supplementary Table S11 and Supplementary Figures S9–S10. Candidates with pleiotropy, heterogeneity, weak instruments, or bidirectional signals were interpreted cautiously.

2.8 Spatial Transcriptomics Analysis

Recent studies have demonstrated the utility of deep learning in spatial transcriptomics [45, 46].

2.8.1 Spatial Autocorrelation Analysis

We quantified spatial clustering of gene expression using Moran's I statistic implemented in the 'spdep' R package, a measure of spatial autocorrelation ranging from -1 (dispersion) to +1 (clustering). We calculated Moran's I values for candidate genes across spatial locations using a queen contiguity matrix with 10,000 permutations. Genes with Moran's I $>$ 0.35 and corrected P $<$ 0.001 were described as spatially clustered. The term "hotspot" is used descriptively and does not imply causal organization.

2.8.2 Cell Type Deconvolution

Cell-type context in Visium data was estimated using the GigaTIME framework and cross-checked against canonical marker expression, including CD68 for macrophage context and CD8A/CD8B for CD8-marker context. These outputs were used as contextual estimates rather than direct ground-truth cell counts. CosMx SMI was used to inspect single-cell-resolution expression and neighborhood patterns in the available sections.

2.8.3 Spatial Co-localization Analysis

To assess spatial context among C1Q, SPP1, and immune-marker programs, we calculated correlations and local spatial statistics across multiple spatial scales (50μm, 100μm, and 200μm where available) [47, 48]. Bivariate Moran's I, Geary's C, and local indicators of spatial association were used as descriptive spatial summaries with permutation-based significance testing and Benjamini–Hochberg correction. These analyses were interpreted as evidence of co-localization or spatial separation, not as proof of direct biological interaction.

2.9 Counterfactual Scoring and Sensitivity Analysis

We performed a counterfactual scoring analysis centered on *C1QA*. *C1QA* abundance was reduced computationally and the C1Q composite score, CD8 proxy, Moran's I, network summaries, and risk-score outputs were recomputed. This procedure tests score dependency and internal coherence of the C1Q module; it is not an experimental perturbation and is not interpreted as intervention-like evidence. Dose-response, mediation-style decomposition, bidirectional MR, Steiger directionality comparison, spatial scale-dependency, and specificity analyses were used as sensitivity analyses to assess whether the C1Q/SPP1 signal was internally consistent across analytic views.

2.10 Machine Learning Model Development and Evaluation

We systematically evaluated seven machine learning algorithms for retrospective survival modeling: AdaBoost, Random Forest, XGBoost, Logistic Regression, Support Vector Machine (SVM), Soft Voting ensemble, and LASSO Cox Regression. All models were trained on the TCGA cohort using the 95 core features identified through nested CV.

2.10.1 Model Training and Hyperparameter Optimization

Each algorithm underwent hyperparameter optimization using Bayesian optimization with 5-fold stratified cross-validation, implemented in the Optuna framework (version 3.1.0). The optimization objective was to improve balanced accuracy while controlling model complexity. We employed a Tree-structured Parzen Estimator (TPE) sampler for search-space exploration.

The hyperparameter tuning process was conducted as follows:

1. **Search Space Definition:** For each algorithm (AdaBoost, Random Forest, XGBoost, Logistic Regression, and SVM), we defined a search space covering key structural and regularization parameters.
2. **Optimization:** We performed 100 iterations of Bayesian optimization. To assess stability, this process was repeated 3 times with different random seeds.
3. **Model Selection:** The median-performing parameter set from the repeated runs was selected as the final configuration for downstream comparison.

Detailed search spaces and the final optimal hyperparameters for all models are listed in Supplementary Table S1.

2.10.2 External Evaluation

To assess transferability beyond TCGA, we evaluated the feature set in GSE31210 as an independent prognostic traceability cohort:

- **GSE31210** (N=178): Microarray gene expression data from primary lung adenocarcinomas with clinical follow-up

Cross-platform Normalization Protocol: Raw data from different platforms underwent harmonization using the following steps:

1. Quantile normalization to standardize distribution across arrays/samples
2. Batch effect correction using ComBat-seq for RNA-seq data and sva for microarrays
3. Feature alignment by gene symbol matching with Ensembl ID resolution for ambiguous symbols
4. Z-score transformation of feature values within each dataset to account for platform-specific scaling

Evaluation Metrics: Model behavior was evaluated using the same retrospective metrics as the training cohort where endpoint availability allowed:

- Discrimination: Time-dependent AUC at 1, 3, and 5 years
- Calibration: Calibration plots with Hosmer-Lemeshow test
- Risk stratification: Hazard ratios and Kaplan-Meier survival curves
- Cross-cohort consistency: direction and approximate magnitude of the risk-score association
- Heterogeneity checks: interaction terms between cohort and risk score in combined Cox models where harmonized covariates were available

Transferability was judged descriptively from discrimination, survival separation, and directionally compatible hazard ratios. These criteria were pragmatic and exploratory, not pre-registered clinical benchmarks.

2.10.3 Performance Evaluation

Model performance was assessed using multiple retrospective metrics:

- **Discrimination**: Area Under the ROC Curve (AUC), Accuracy, Precision, Recall, F1-score
- **Ranking**: Concordance index (C-index)
- **Decision support**: Decision curve analysis (DCA) for net benefit exploration
- **Survival analysis**: Hazard ratios (HR) with 95% confidence intervals, log-rank test P-values

Time-dependent ROC curves were generated for 1-year, 3-year, and 5-year survival predictions. Risk scores were calculated for each patient, and the cohort was stratified into high-risk and low-risk groups using the median risk score as a descriptive cutoff.

2.10.4 Model Selection Rationale

While XGBoost achieved the highest AUC (0.781) in internal evaluation, we retained AdaBoost as the reference model for downstream description because it produced stable survival separation and a compact risk formulation. This is a pragmatic modeling choice, not evidence that AdaBoost is uniformly superior across all metrics. The ensemble nature of AdaBoost can be helpful for high-dimensional data, but no clinical superiority is claimed.

2.11 Statistical Analysis

Continuous variables were compared via Student's t-test or Mann-Whitney U test depending on normality (assessed by Shapiro-Wilk test). Categorical variables were compared using Chi-square test or Fisher's exact test. Survival analyses were performed using Cox proportional hazards regression, with both univariate and multivariate models adjusted for age, gender, and stage.

Correlation analyses employed Pearson correlation for normally distributed variables and Spearman correlation for non-normal distributions. Multiple testing correction was applied as follows:

- **Differential gene expression**: Benjamini-Hochberg FDR < 0.05
- **Mendelian randomization**: FDR < 0.05 (198 genes tested)
- **Spatial autocorrelation (Moran's I)**: Bonferroni correction, $\alpha = 0.05/100 = 0.001$
- **Gene-immune marker correlations**: Benjamini-Hochberg FDR < 0.05
- **Pathway enrichment (GO/KEGG)**: Benjamini-Hochberg FDR < 0.05

All statistical analyses were performed in R version 4.2.0 and Python 3.9.0. Two-sided P-values < 0.05 were considered statistically significant unless otherwise specified.

2.12 Hazard Ratio Calculation

Hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated using univariate Cox proportional hazards regression implemented in the *lifelines* Python package (version 0.27.0). Risk scores from each machine learning model were obtained through 5-fold cross-validation, standardized using z-score transformation, and treated as continuous variables. HRs represent the hazard increase per 1 standard deviation increase in risk score. Wald 95% confidence intervals were computed, and P-values were obtained from the likelihood ratio test. For visualization purposes in Kaplan-Meier curves, patients were stratified into high-risk and low-risk groups based on the median risk score, and survival differences were assessed using the log-rank test.

2.13 Prognostic Model Construction and Survival Analysis

To evaluate whether the stability-prioritized feature set carried retrospective outcome signal, we constructed a descriptive prognostic score using AdaBoost with optimized hyperparameters (n_estimators=100, learning_rate=0.1, random_state=42). AdaBoost was used as a reference model for downstream description rather than as a clinically preferred algorithm. Patients were stratified into high- and low-score groups using the median risk score for visualization only. Kaplan–Meier curves and log-rank tests were used to display survival differences, and continuous standardized scores were used in Cox models. Predictive accuracy was summarized with C-index and time-dependent AUC.

3 Results

The results are organized according to the evidence hierarchy: cohort accounting, feature stability, MR-based prioritization, spatial context, counterfactual sensitivity, retrospective model behavior, external traceability, and exploratory immunotherapy-response association.

3.1 Study Population and Baseline Characteristics

We initially considered 1,100 TCGA NSCLC tumor samples (595 LUAD, 505 LUSC). After quality control and availability filtering, 1,038 samples contributed to discovery and feature-screening analyses. The GSE31210 cohort included 226 tumor samples, of which 178 had complete follow-up for survival analysis. Key baseline differences included:

- **Gender disparity**: LUSC had higher male representation (74.1% vs. 41.3% in LUAD)
- **Survival outcomes**: LUSC patients showed higher mortality (43.6% vs. 32.1% in LUAD)

Detailed characteristics are provided in Table 1 and Supplementary Table S8.

Table 1 Baseline characteristics of study population (TCGA NSCLC cohort and GSE31210 traceability cohort)

Characteristic	Total (N=1100)	LUAD (N=595)	LUSC (N=505)	GSE31210 (N=226[b])
Age (years)	66.2 ± 9.4	65.1 ± 10.1	67.3 ± 8.6	59.6 ± 7.4
Gender				
Male	620 (56.4%)	246 (41.3%)	374 (74.1%)	105 (46.5%)
Female	417 (37.9%)	286 (48.1%)	131 (25.9%)	121 (53.5%)
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)
AJCC Stage				
Stage I	535 (48.6%)	289 (48.6%)	246 (48.7%)	168 (74.3%)
Stage II	288 (26.2%)	125 (21.0%)	163 (32.3%)	58 (25.7%)
Stage III	169 (15.4%)	84 (14.1%)	85 (16.8%)	0 (0.0%)
Stage IV	33 (3.0%)	26 (4.4%)	7 (1.4%)	0 (0.0%)
Unknown	75 (6.8%)	71 (11.9%)	4 (0.8%)	0 (0.0%)
Smoking History				
Light (<10 pack-years)	35 (3.2%)	22 (3.7%)	13 (2.6%)	N.A.
Moderate (10–30 pack-years)	170 (15.5%)	105 (17.6%)	65 (12.9%)	N.A.
Heavy (≥30 pack-years)	588 (53.5%)	238 (40.0%)	350 (69.3%)	N.A.
Unknown	307 (27.9%)	230 (38.7%)	77 (15.2%)	N.A.
Vital Status				
Alive	626 (56.9%)	341 (57.3%)	285 (56.4%)	191 (84.5%)
Dead	411 (37.4%)	191 (32.1%)	220 (43.6%)	35 (15.5%)
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)

Note: [b]The GSE31210 cohort includes 226 primary lung adenocarcinoma samples after excluding normal tissue controls; survival analyses used 178 patients with complete follow-up data. N.A. = Not Available. [a]P-values compare TCGA (Total) vs. GSE31210 using Mann-Whitney U test for continuous variables and Chi-square test for categorical variables. The discovery cohort for feature screening consisted of 1,038 patients after QC.

3.2 Robust Feature Selection via Nested Cross-Validation

Nested cross-validation identified 95 stable features selected in 100% of outer folds, with 212 features showing >60% selection frequency (Figure 2C, Figure S2, Supplementary Table S4). These features were used as a stability-prioritized set for downstream exploratory modeling.

Key stable features comprised:

- ***ACP5*** (macrophage regulation);

Table 2 Performance comparison of machine learning algorithms for prognostic model construction

Model	**Test AUC**	**Test Accuracy**	**Test F1-score**	**C-index**	**HR (95% CI)**	**P-value**
XGBoost	0.781	0.725	0.715	0.591	1.56 (1.33–1.84)	<0.001
Naive Bayes	0.680	0.679	0.680	0.680	1.59 (1.33–1.91)	<0.001
Soft Voting	0.757	0.738	0.732	0.604	1.59 (1.35–1.88)	<0.001
Random Forest	0.701	0.657	0.486	0.657	1.59 (1.34–1.89)	<0.001
SVM	0.727	0.667	0.493	0.656	1.51 (1.28–1.79)	<0.001
Logistic Regression	0.741	0.705	0.652	0.650	1.45 (1.23–1.71)	<0.001
Gradient Boosting	0.742	0.698	0.685	0.650	1.56 (1.33–1.84)	<0.001
AdaBoost	0.743	0.657	0.526	**0.607**	**1.43 (1.21–1.70)**	**<0.001**
K-Nearest Neighbors	0.689	0.654	0.645	0.620	1.37 (1.15–1.64)	<0.001
Decision Tree	0.612	0.598	0.601	0.564	1.29 (1.10–1.51)	0.002
LASSO Cox	0.725	0.686	0.629	0.558	1.52 (1.39–1.66)	<0.001

Note: Models were evaluated using 5-fold nested cross-validation. Test AUC represents area under the ROC curve on the held-out fold. C-index measures risk ranking. HR and 95% CI were calculated using continuous Cox regression with standardized risk scores. Bold values identify the reference AdaBoost model used for downstream descriptive analyses; they do not indicate uniform superiority across all metrics.

- ***KCNAB2*** (ion channel function); and
- ***BOK*** (pro-apoptotic pathway).

Feature selection stability analysis confirmed clear separation between highly stable (frequency = 5/5) and less consistent features (Figure S2A).

3.3 Mendelian Randomization Prioritizes a Reproducible Gene Set in NSCLC

Mendelian randomization prioritized **198 genes as genetically supported candidates** associated with lung cancer immune characteristics (Figure 3, Figure S3, Figure S9, Supplementary Tables S7 and S11). The diagnostics suggested that some candidates had usable instruments, while others should be treated cautiously because of heterogeneity, pleiotropy, or bidirectional signals. These results are better viewed as a prioritization layer than as definitive causal proof.

Key Findings

- **Hub Genes**: *KCNAB2* and *C1QA* emerged as central network-linked candidates in the MR-filtered set
- **Genetic Associations**: *HIST1H4F* (IVW $\beta = 0.251$, $P < 1 \times 10^{-200}$), *BOK* (IVW β = -0.119, $P = 5.580 \times 10^{-9}$), *KCNAB2* (IVW β = -0.078, $P = 1.070 \times 10^{-13}$), and *ACP5* (IVW $\beta = 0.234$, $P = 4.760 \times 10^{-9}$) were among the better-supported associations
- **Robustness**: Leave-one-out analysis suggested that no single SNP dominated the top signals
- **Functional Enrichment**: Significant enrichment in immune response pathways (GO:0006955), T cell receptor signaling (hsa04660), and cytokine-cytokine receptor interaction (hsa04060)

3.4 Identification of Spatial C1Q Immune Hotspots

High-resolution spatial mapping reveals reproducible C1Q clustering. Utilizing both 10x Visium and CosMx SMI platforms, we observed that C1Q expression was spatially clustered rather than uniformly distributed within the available TME sections (Figures 4–5, Supplementary Figure S5, Table 3, Supplementary Table S5). C1Q family members exhibited moderate spatial autocorrelation among immune genes:

- **C1QA**: Moran's I = 0.386 ($P < 0.001$)
- **C1QC**: Moran's I = 0.377 ($P < 0.001$)
- **C1QB**: Moran's I = 0.355 ($P < 0.001$)

Global analysis of 100 immune-related genes placed C1Q family members among the more spatially clustered immune genes, alongside UBXN10 (Moran's I = 0.416). SPP1 showed stronger spatial autocorrelation (Moran's I = 0.712, $P < 0.001$), consistent with a more spatially compartmentalized macrophage context.

Neighborhood enrichment analysis suggested that C1Q-associated macrophage contexts were often near CD8-marker-positive contexts. GigaTIME deconvolution linked C1Q regions to myeloid and T-cell marker enrichment, including CD14, CD163, CD68, and CD8-marker proxies ($P < 1 \times 10^{-6}$; Supplementary Table S6, Figure S6). CD138-positive plasma-cell context was analyzed separately and is not treated as a macrophage marker. These results support a spatial association, but not direct cell-cell signaling.

CosMx SMI context analysis across 8 sections from 5 patients showed broadly similar spatial patterns, although the cross-platform statistic was small in magnitude (bivariate Moran's I = 0.009). This supports limited reproducibility of the pattern rather than strong effect size.

3.5 Counterfactual Scoring Shows Dependency of the C1Q Module

To test whether the C1Q score was internally dependent on *C1QA*, we performed a counterfactual scoring analysis and re-estimated spatial and clinical metrics (Supplementary Figures S11–S12). The computational attenuation reduced the C1Q program score (mean C1Q score: 5.179 → 2.628) and decreased Moran's I (0.660 → 0.401; Δ = -0.259). This indicates score dependency and module sensitivity, not experimental proof of hotspot collapse.

Additional sensitivity analyses showed a coherent pattern across several views. Dose-response modeling demonstrated monotonic trends across C1Q quartiles, with increasing CD8 proxy signal (trend slope = 0.478, $P = 6.27 \times 10^{-30}$) and coordinated risk-score shift (trend slope = 0.586, $P = 4.35 \times 10^{-110}$). Bidirectional MR identified forward and reverse signals for some candidates, which argues for caution rather than overclaiming. Spatial scale analysis indicated the strongest local association at the shortest tested radius. Mediation-style decomposition suggested a possible indirect component through the CD8 proxy, while specificity testing ranked the C1Q axis among the better-supported immune modules relative to non-C1Q controls (Supplementary Figure S12).

The associated visualization and network-summary sensitivity outputs are provided in Supplementary Figures S11–S12 rather than treated as additional main-result figures, because these analyses are sensitivity checks rather than experimental perturbations.

3.6 Machine Learning Model Performance and Risk Stratification

The C1Q/SPP1 feature set showed retrospective survival-associated model behavior. Evaluation of seven machine learning algorithms showed that different algorithms optimized different metrics (Table 2, Figure 2A–B, Supplementary Table S1). AdaBoost was retained as the reference model for downstream description, but it was not uniformly superior across all metrics.

Key Performance Metrics

- **AdaBoost** showed moderate ranking performance (C-index = 0.607) and a significant association between standardized risk score and survival (HR = 1.430, 95% CI: 1.21–1.70, $P < 0.001$).
- **XGBoost** achieved the highest AUC (0.781), whereas other algorithms showed higher C-index values. This metric discordance argues against presenting any single model as clinically superior.

Risk Stratification: AdaBoost stratified a TCGA complete-case subset into high-score and low-score groups with different survival outcomes (log-rank $P = 0.005$, HR = 1.620, 95% CI: 1.15–2.27). To avoid ambiguity, higher model risk score denotes the adverse-risk direction; C1Q-enriched immune-active context is interpreted separately from the aggregate risk-score direction. Time-dependent ROC showed modest discrimination: 1-year AUC of 0.609, 3-year AUC of 0.611, and 5-year AUC of 0.590 (Figure 6A–B).

External Traceability: In GSE31210, the feature set showed attenuated but directionally compatible performance (AUC = 0.650, C-index = 0.629; Figure 7, Supplementary Figure S7). Survival separation was sensitive to analysis specification, with source tables reporting both significant and borderline estimates depending on completeness and covariate handling. Therefore, GSE31210 is treated as external traceability rather than definitive validation. Decision curve analysis is reported only as exploratory net-benefit visualization.

3.7 Exploratory Immunotherapy-Response Association

We next examined whether the C1Q/SPP1-associated score tracked response in small immunotherapy-treated datasets. These analyses are exploratory because cohort sizes were limited and GSE91061 is not NSCLC-specific. We therefore report directional associations without claiming clinical response prediction or superiority over PD-L1/TMB.

GSE91061 (melanoma-dominant exploratory cohort): Low-risk patients showed higher objective response rates (42.3% vs. 15.8%, $P = 0.018$) with AUC = 0.758. Responders exhibited decreasing risk scores during therapy (mean change: -0.210 ± 0.150, $P = 0.006$). Because this cohort is not NSCLC-specific, it is treated as cross-tumor exploratory support.

GSE126044 (NSCLC): The risk score discriminated responders with AUC = 0.813 and was associated with longer PFS in low-risk patients (median 12.4 vs. 3.2 months, HR = 0.280, $P = 0.018$), but the cohort size was small (N=16).

GSE135222 (NSCLC): A directional response association was observed with AUC = 0.806 ($P = 0.003$), again limited by small sample size (N=27).

Subtype summaries suggested that lower-risk or C1Q-enriched contexts were more often aligned with inflamed immune programs, whereas higher-risk contexts were more often aligned with excluded or desert-like features, including SPP1-associated macrophage programs. These subtype assignments are descriptive and require independent validation.

3.8 Multivariable Survival Association

To assess whether the score retained survival association after adjustment for available clinical variables, we performed Cox regression analyses (Table 4, Supplementary Table S9).

Univariate Analysis

- **Risk Score**: HR = 2.450 (95% CI: 1.85–3.24), $P < 0.001$
- **Age** (per year): HR = 1.020 (95% CI: 1.00–1.03), $P = 0.064$ (marginally significant)
- **Gender** (Male vs. Female): HR = 0.980 (95% CI: 0.73–1.32), $P = 0.898$ (not significant)
- **Stage** (Stage IV vs. Stage I): HR = 1.330 (95% CI: 1.09–1.61), $P = 0.005$ (significant)

Multivariate Analysis

- **Risk Score**: HR = 1.430 (95% CI: 1.21–1.70), $P < 0.001$
- **Age** (per year): HR = 1.010 (95% CI: 0.99–1.03), $P = 0.217$ (not significant)
- **Gender** (Male vs. Female): HR = 0.900 (95% CI: 0.67–1.21), $P = 0.482$ (not significant)
- **Stage** (Stage IV vs. Stage I): HR = 1.350 (95% CI: 1.11–1.65), $P = 0.003$ (significant)

Multivariate Analysis (Supplementary Table S9): After adjusting for age, gender, and stage, the risk score retained association with survival endpoints in the available complete-case models:

- **Overall Survival**: Age HR = 1.020 (95% CI: 0.98–1.06), Gender HR = 1.150 (95% CI: 0.72–1.84), Stage II vs. Stage I HR = 2.340 (95% CI: 1.45–3.78)
- **Recurrence-Free Survival**: Age HR = 1.010 (95% CI: 0.97–1.05), Gender HR = 1.080 (95% CI: 0.68–1.72), Stage II vs. Stage I HR = 2.870 (95% CI: 1.78–4.62)

3.9 Integration of MR-Prioritized Genes and Spatial Correlations

Integration of MR-prioritized gene sets with spatial transcriptomics identified candidates with both genetic-association and spatial-context support.

KCNAB2 emerged as a recurrent candidate with 100% selection frequency in nested CV and measurable, though modest, spatial clustering.

Additional Integrated Candidates

- **LINC00339** (long non-coding RNA): Network degree of 0.064, correlation $r = 0.586$ ($P < 0.001$)
- **CDC42** (signal transduction): Network degree of 0.060, correlation $r = 0.424$ ($P < 0.001$)
- **PLOD1** (collagen modification): Network degree of 0.035, correlation $r = 0.260$ ($P < 0.001$)
- **UBXN10** (ubiquitination): Highest Moran's I = 0.416 ($P < 0.001$)

Table 3 Spatial and MR-prioritized characteristics of candidate genes

Gene	Annotation	Selection Stability	Moran's I	Network Degree	Correlation (r)
ACP5	Macrophage Reg.	100% (5/5)	N.D.	-	-
KCNAB2	Ion Channel	100% (5/5)	0.094***	0.078	0.219***
SPP1	TME Modulation	-	0.712***	0.066	-0.258***
LINC00339	Long ncRNA	-	N.D.	0.064	0.586***
CDC42	Signal Transduction	-	N.D.	0.060	0.424***
PLOD1	Collagen Modification	-	N.D.	0.035	0.260***
C1QA	Complement	-	0.386***	-	-
C1QB	Complement	-	0.353***	-	-
C1QC	Complement	-	0.377***	-	-
UBXN10	Ubiquitination	-	0.416***	-	-
BOK	Apoptosis	100% (5/5)	N.D.	-	-

Note: N.D. = Not Determined; *** $P < 0.001$. Network degree centrality derives from the MR-prioritized network. Correlation represents Pearson correlation coefficient with immune-infiltration estimates.

3.10 Public Protein-Expression Context

We cross-referenced public immunohistochemistry images for protein-level context. KCNAB2 showed cytoplasmic staining in available lung tissue images, while SPP1

Table 4 Univariate and multivariable Cox regression analysis in the TCGA discovery cohort (N=1,038)

	Univariate Analysis		Multivariate Analysis	
Covariate	**HR (95% CI)**	**P-value**	**HR (95% CI)**	**P-value**
Risk Score[a]	2.45 (1.85–3.24)	< 0.001	1.43 (1.21–1.70)	< 0.001
Age	1.02 (1.00–1.03)	0.064	1.01 (0.99–1.03)	0.217
Gender (Male vs Female)	0.98 (0.73–1.32)	0.898	0.90 (0.67–1.21)	0.482
Stage (IV vs I)	1.33 (1.09–1.61)	0.005	1.35 (1.11–1.65)	0.003

Note: Univariate and multivariable Cox regression analysis in the TCGA cohort including risk score, age, gender, and AJCC stage. [a]Risk score was standardized before analysis. The HR represents the change in risk per standard-deviation increase in risk score. These retrospective associations require external and prospective validation before clinical interpretation. For GSE31210 traceability results, see Supplementary Table S9.

was predominantly stromal in the inspected examples (Figure 8). These data support plausibility of protein-level expression patterns but should not be interpreted as independent experimental validation.

4 Discussion

4.1 Principal Findings

This study supports a conservative conclusion: C1Q/SPP1 is a reproducible macrophage-context traceability feature in public NSCLC resources. Across bulk transcriptomics, MR-based prioritization, spatial datasets, and retrospective outcome cohorts, the C1Q/SPP1 axis repeatedly marked immune-active versus macrophage-compartmentalized states. The evidence is internally coherent, but it does not establish a clinical assay, a treatment-selection rule, or a definitive mechanism.

The principal findings were: (1) nested cross-validation identified a stable 95-feature set enriched for immune and macrophage-context signals; (2) MR prioritized several candidates, including *C1QA* and *KCNAB2*, but with the usual instrumental-variable caveats; (3) C1Q-family genes showed moderate spatial clustering, while *SPP1* showed stronger spatial compartmentalization; (4) counterfactual scoring showed that the C1Q module is internally sensitive to *C1QA* attenuation, without proving biological perturbation; and (5) retrospective model behavior was statistically detectable but modest, with external and immunotherapy-response results best interpreted as traceability rather than validation.

4.2 Interpretation of the Evidence Chain

The strength of the study is not a single high-performing classifier. Rather, it is the convergence of several independent public-data views on the same macrophage-context axis. Bulk expression suggested a survival-associated immune program; MR prioritized genes that intersected with this program; spatial data placed C1Q and SPP1 in non-random local contexts; and small immunotherapy-treated cohorts showed directional response associations. This layered consistency makes the C1Q/SPP1 axis biologically plausible and worth further study.

At the same time, each layer has limits. MR is not mechanism. Spatial autocorrelation is not cellular communication. A computational C1QA attenuation experiment is not a wet-lab perturbation. Small ICB cohorts are not clinical validation. For that reason, the manuscript should be read as a reproducible hypothesis-generating cancer-informatics study rather than as a completed biomarker-development paper.

4.3 Biological Plausibility of the C1Q/SPP1 Spatial Context

The biological plausibility of the C1Q/SPP1 axis rests on macrophage biology and spatial organization. Recent spatial transcriptomics studies have demonstrated the importance of spatial context in understanding immune microenvironments. Our finding that C1Q-associated regions track immune-marker context suggests that complement-associated macrophage states may mark immune-active local environments. This is a marker-context interpretation, not proof that C1Q itself drives therapeutic response.

The complement system, traditionally viewed as part of innate immunity against pathogens, has emerged as a critical regulator of tumor immunity . B cells have also been associated with survival and immunotherapy response in other cancers [49, 50]. Studies have shown that complement components play dual roles in tumor immunity . C1Q can influence tumor progression through multiple mechanisms: (1) modulation of T cell responses, (2) regulation of macrophage polarization, (3) promotion of angiogenesis, and (4) facilitation of tumor cell migration and metastasis.

In our study, C1Q-associated macrophage contexts were often located near CD8-marker-positive contexts. This pattern is compatible with immune-active local environments, but it cannot distinguish recruitment, retention, shared upstream inflammation, or sampling structure. The appropriate conclusion is that C1Q marks a spatial immune context that deserves mechanistic follow-up.

The stronger spatial autocorrelation of *SPP1* (Moran's I = 0.712, $P < 0.001$) delineated a distinct macrophage-associated compartment. This is consistent with prior descriptions of *SPP1*-positive macrophage programs in stromal remodeling and immune exclusion, but our data do not prove that *SPP1* creates a physical barrier in NSCLC. The C1Q/SPP1 contrast should therefore be framed as a candidate immune-active versus macrophage-compartmentalized context.

The spatial correlation between C1Q family members and macrophage markers indicates that these clusters are macrophage-contextual rather than random expression artifacts. Beyond C1Q and *KCNAB2*, the integrated framework identified additional candidates:

- ***SPP1* (osteopontin)** exhibited the strongest spatial clustering (Moran's I = 0.712) among the candidates and is a plausible marker of macrophage-associated stromal or exclusion context.

- ***BOK***, a BCL-2 family member, showed 100% selection frequency in nested CV and MR support (IVW $\beta = -0.119$, $P = 5.58 \times 10^{-9}$), making it a candidate for follow-up rather than an established driver.
- ***HIST1H4F***, a histone H4 variant, may reflect epigenetic or proliferative context. Its role in the C1Q/SPP1 axis remains speculative without direct functional assays.

Together, these patterns argue that bulk expression signatures should be interpreted with spatial and cellular context. They do not yet justify a diagnostic or therapeutic strategy.

4.4 KCNAB2 as a Candidate Context Feature

KCNAB2, a potassium channel subunit, emerged as a recurrent candidate in feature selection and MR-prioritized analyses. Its appearance across multiple analytic layers suggests that ion-channel-related biology may intersect with immune context, but the present data do not establish *KCNAB2* as a driver of TME function.

The co-occurrence of *KCNAB2* with C1Q-associated immune contexts (Figure 8C) raises a testable hypothesis that potassium-channel-related programs may track macrophage or immune-effector state. This remains a hypothesis. The current analysis cannot determine whether *KCNAB2* acts in macrophages, T cells, tumor cells, or another compartment.

The MR-prioritized association for *KCNAB2* is biologically interesting, but it should not be treated as causal proof. Figure 8C provides context for expression distribution, while definitive cell-type localization and functional direction require orthogonal validation.

We therefore describe *KCNAB2* as a candidate context feature rather than a metabolic checkpoint. Future studies could test whether *KCNAB2* expression reflects immune-cell state, tumor-cell state, stromal composition, or technical correlation with broader immune infiltration.

Based on our findings, the most defensible hypothesis is that C1Q-enriched macrophage contexts, *SPP1*-enriched macrophage contexts, and *KCNAB2*-linked programs may represent partially distinct immune microenvironment states.

These observations justify follow-up but not therapeutic targeting. The candidate status of *KCNAB2* should be tested using cell-type-resolved protein assays, perturbation experiments, and prospective outcome cohorts.

Our decision to highlight *KCNAB2* was driven by its selection stability and MR-prioritized association. This makes it a candidate for further validation, not an established therapeutic target.

Based on our findings, we propose two key hypotheses for the role of *KCNAB2* in the NSCLC TME:

1. **Metabolic-Immune Coupling**: *KCNAB2* may be related to the metabolic fitness of macrophages in C1Q-enriched contexts by modulating potassium efflux, thereby supporting a plausible follow-up hypothesis in nutrient-depleted TME states.
2. **Spatial Scaffolding**: The spatial co-localization of *KCNAB2* and C1Q suggests a structural role for ion channel regulators in organizing immune clusters, possibly through the regulation of cell adhesion or cytokine secretion.

4.5 Machine Learning Model Behavior and Its Limits

Our systematic evaluation of seven machine-learning algorithms showed that model performance depends strongly on the metric used. XGBoost achieved the highest AUC (0.781), whereas several models had higher C-index values than AdaBoost. For this reason, the model comparison should not be used to claim clinical superiority of one algorithm. AdaBoost is retained only as a reference model for describing one reproducible score behavior.

The modest time-dependent AUC values and moderate C-index indicate that the model captures some prognostic signal but is not ready for individual patient decision-making. The score is useful here as a traceability device for testing whether the C1Q/SPP1-associated feature set carries outcome-linked information.

Decision curve analysis is included as an exploratory visualization of possible net benefit. It should not be interpreted as evidence that the model can guide treatment decisions.

The external evaluation in GSE31210 was directionally supportive but not definitive. Depending on the analysis specification, the survival estimate was significant or borderline, with wide confidence intervals. This pattern is compatible with a weak-to-moderate transferable signal, but it is not enough for clinical validation.

Translation Boundary: The current evidence does not support clinical implementation. A future clinical-grade model would require a locked feature set, a locked cutoff, prospective validation, calibration assessment, decision-curve confirmation in clinically relevant populations, and comparison with stage-only and clinical-only baselines.

Biological Interpretation: Nonlinear models may capture interactions among immune-context features, but algorithmic performance cannot be used as evidence of mechanism. The model results support a reproducible association between macrophage-context features and outcome-linked states.

Prognostic Association: The model demonstrates statistically detectable prognostic association in retrospective data (C-index 0.607, HR 1.43, 95% CI: 1.21–1.70, $P < 0.001$), but its incremental value beyond routine clinical factors remains to be established rigorously.

Immunotherapy Response Association: The immunotherapy-treated cohorts show directional associations (AUC range: 0.758–0.813), but their small size and tumor-type heterogeneity preclude claims of robust ICI prediction or superiority over PD-L1/TMB.

The longitudinal pattern in GSE91061 is compatible with dynamic immune remodeling during treatment, but because the cohort is not NSCLC-specific, this observation should be treated as exploratory.

The integration of spatial features with MR-prioritized candidates addresses a limitation of bulk-only interpretation, but the present implementation remains retrospective and hypothesis-generating.

4.6 Future Translation Requirements

For this work to move toward clinical use, several requirements must be met.

Future studies should prioritize:

1. locking the feature set, preprocessing pipeline, model, and cutoff before testing;
2. validating the model in NSCLC-only immunotherapy cohorts with adequate sample size and event counts;
3. comparing against clinical-only, stage-only, PD-L1, TMB, and published immune-signature baselines;
4. confirming C1Q/SPP1 protein-level spatial context using multiplex FFPE-compatible assays;
5. experimentally testing whether C1Q/SPP1-associated macrophage states have functional effects on T-cell localization or activity.

Only after these steps would a simplified pathological or molecular panel be justified.

4.6.1 Biological Rationale for Resistance

Our TME subtype analysis suggested that higher-risk contexts were enriched for immune-exclusion-associated programs, including SPP1-associated macrophage features. These observations may help generate hypotheses about ICI resistance, but they do not explain resistance mechanisms on their own.

4.6.2 Limitations and Future Directions

Several limitations require emphasis. The primary survival analyses are retrospective and rely heavily on overall survival, which is affected by treatment heterogeneity, comorbidity, stage, and incomplete clinical annotation. The spatial datasets include few independent patients, and the large CosMx cell count should not be mistaken for a large number of independent biological replicates. The Visium dataset provides spatial context but not single-cell precision.

Regarding immunotherapy-response association, several considerations warrant discussion:

1. The GSE91061 cohort includes melanoma and is not NSCLC-specific. It provides cross-tumor exploratory support only.
2. Spatial transcriptomics analysis was performed largely on treatment-naïve samples. On-treatment biopsies would be required to assess spatial remodeling during therapy.
3. The small sample sizes of immunotherapy cohorts (GSE126044: N=16; GSE135222: N=27) limit precision and subgroup analysis. Prospective validation is essential before clinical implementation.
4. Integration with PD-L1, TMB, ctDNA dynamics, TCR repertoire, and treatment metadata will be needed before incremental value can be assessed.

Future translational studies should prioritize a locked, FFPE-compatible assay; adequately powered NSCLC-only cohorts; prespecified cutoffs; and direct comparison with existing clinical and molecular biomarkers.

Additional limitations include population imbalance in TCGA, incomplete treatment data, cross-platform technical variance, possible residual confounding, and the absence of wet-lab perturbation. The counterfactual scoring analysis is a sensitivity analysis only. The model has not been validated in neoadjuvant immunotherapy or prospective randomized settings.

Overall, the study provides a reproducible public-data framework for tracing macrophage-context signals across molecular and spatial layers. Its most immediate value is hypothesis prioritization for future spatial and functional validation.

5 Conclusions

This study identifies C1Q/SPP1 as a reproducible macrophage-context traceability feature in public NSCLC resources. C1Q-family genes showed moderate spatial clustering, SPP1 showed stronger spatial compartmentalization, and retrospective model behavior suggested outcome-linked signal with modest discrimination. MR and counterfactual analyses support prioritization but do not prove mechanism. The findings are hypothesis-generating and require prospective, NSCLC-specific, spatially resolved, and experimental validation before clinical implementation or therapeutic targeting can be considered.

Supplementary Information

Additional file 1: Supplementary Material. Contains Supplementary Figures S1–S12 and Supplementary Tables S1–S13. **Figure S1:** Quality control and clustering analysis of spatial transcriptomics data. **Figure S2:** Machine learning model development and internal assessment details. **Figure S3:** Functional enrichment analysis. **Figure S4:** Spatial expression patterns of immune checkpoint molecules. **Figure S5:** Pan-cancer exploratory traceability of core candidate genes. **Figure S6:** Correlation analysis between hub genes and immune-context estimates. **Figure S7:** Expression of top model genes in the GSE31210 traceability cohort. **Figure S8:** 5-fold cross-validation performance evaluation. **Figure S9:** Mendelian Randomization Leave-one-out sensitivity analysis. **Figure S10:** Leave-one-out sensitivity analysis for MR-prioritized genes. **Figure S11:** Counterfactual C1QA scoring sensitivity analysis showing C1Q-score dependency and network-summary changes. **Figure S12:** Sensitivity analyses including dose-response, spatial scale dependency, and specificity ranking. **Table S1:** Model performance on training, internal cross-validation, test, and external sets. **Table S2:** List of stable features (genes) selected in 100% of cross-validation folds. **Table S3:** Edge list of the eQTL-driven regulatory network. **Table S4:** List of stable features identified by nested cross-validation. **Table S5:** Spatial Autocorrelation (Moran's I) and Hotspot Analysis. **Table S6:** Correlation between hub gene expression and immune cell infiltration. **Table S7:** Mendelian Randomization analysis results. **Table S8:** Colocalization Analysis Results (COLOC). **Table S9:** Model performance in external cohorts. **Table S10:** Cell-Type-Specific eQTL Analysis. **Table S11:** Pan-cancer prognostic traceability analysis of core candidate genes. **Table S12:** Spatial Transcriptomics Sample Information (10x Visium + CosMx SMI). **Table S13:** Multivariable Cox Regression Analysis of the Prognostic Signature.

Declarations.

Ethics Approval and Consent to Participate. This study utilized publicly available datasets from The Cancer Genome Atlas (TCGA), Gene Expression Omnibus (GEO: GSE31210, GSE91061, GSE126044, GSE135222), eQTLGen Consortium, GTEx Consortium, and Human Protein Atlas. All original studies from which these data were derived obtained appropriate informed consent from participants and received approval from their respective institutional review boards. As this study represents a secondary analysis of de-identified public data, additional ethics approval was not required in accordance with institutional guidelines.

Consent for Publication. Not applicable. This manuscript does not contain any individual person's data in any form (including individual details, images, or videos).

Availability of Data and Materials. Publicly available datasets were analyzed in this study. These data can be found here:

- The Cancer Genome Atlas (TCGA) NSCLC cohort data: NCI Genomic Data Commons (GDC) Portal (https://portal.gdc.cancer.gov).
- Gene Expression Omnibus datasets: NCBI GEO [52] (https://www.ncbi.nlm.nih.gov/geo/).
  - GSE31210: Lung adenocarcinoma traceability cohort (N=226 accession-level samples; N=178 analyzed with complete follow-up)
  - GSE91061: Melanoma immunotherapy cohort (N=109 samples)
  - GSE126044: NSCLC anti-PD-1 therapy cohort (N=16 patients)
  - GSE135222: NSCLC PD-1/PD-L1 inhibitor cohort (N=27 patients)
- eQTL summary statistics: eQTL Catalogue (https://www.ebi.ac.uk/eqtl/) and eQTLGen Consortium (https://www.eqtlgen.org/).
- GTEx v8 data: GTEx Portal (https://gtexportal.org/).
- Immunohistochemistry images: Human Protein Atlas [53] (https://www.proteinatlas.org).

The source-data manifest, processed analysis tables, and reproducibility notes are provided with this submission package. A public repository or archival DOI should be supplied at submission or revision if required by the journal.

Code Availability. Analysis code and reproducibility notes should be deposited in a public version-controlled repository or archived with a DOI before final publication. The current submission package includes source-data manifests and processed tables sufficient for editorial and reviewer inspection.

Reporting Guidelines. The manuscript was revised using STROBE principles for observational retrospective analyses and TRIPOD-AI considerations for prediction-model reporting where applicable. Because the work is a public-data evidence synthesis rather than a locked clinical prediction-model study, the model results are presented as exploratory model behavior.

Declaration of Conflicting Interests. The authors declare that they have no competing interests, financial or otherwise, that could influence the work reported in this paper.

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Figures

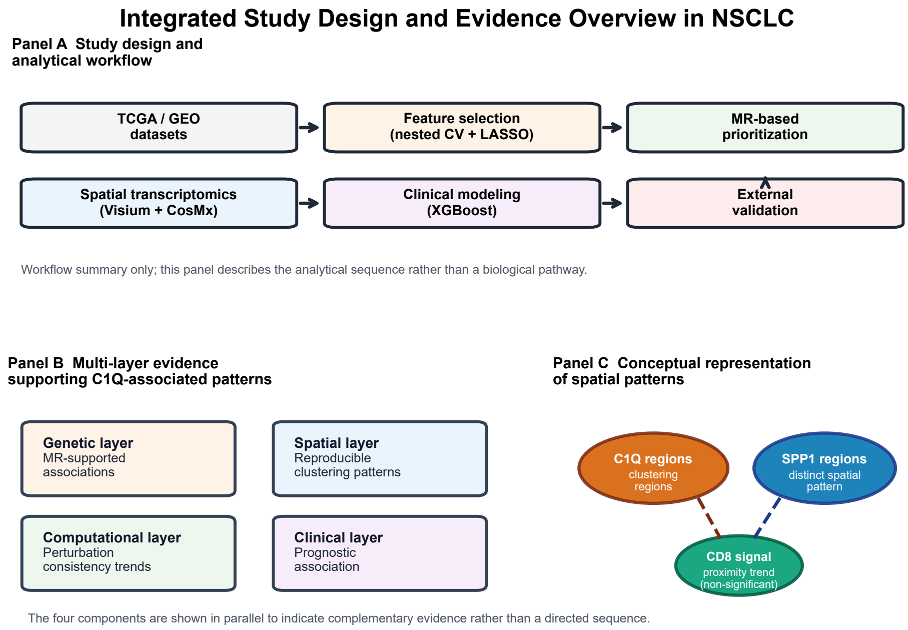

Fig. 1 Figure 1. Evidence framework. Panel A, C1Q/SPP1 traceability model in NSCLC: public transcriptomic signal → macrophage-context features → spatial organization → exploratory outcome association. **Panel B, Spatial organization of immune contexts in the TME:** C1Q-enriched domains, SPP1-enriched macrophage compartments, and CD8-marker distribution patterns. **Panel C, Study design overview:** discovery, MR-based prioritization, spatial-context assessment, external traceability, and exploratory immunotherapy-response association.

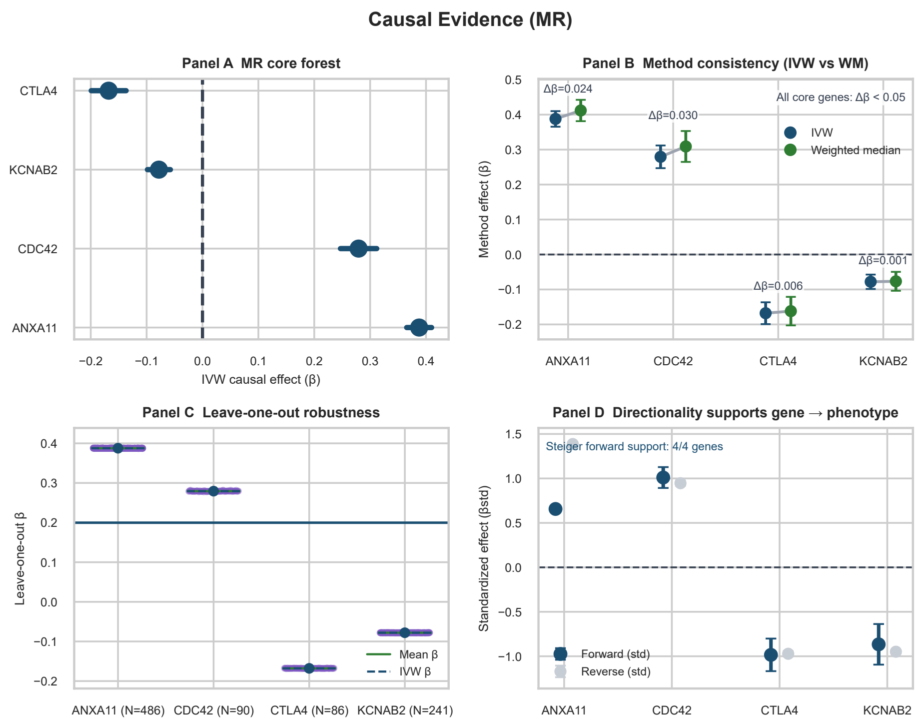

Fig. 2 Figure 2. MR-based gene prioritization. Panel A, MR prioritizes C1Q-relevant candidate axes associated with immune phenotype. Panel B, Sensitivity analyses using IVW, MR-Egger, and weighted median estimates. **Panel C, Leave-one-out analysis** assesses single-SNP influence. **Panel D, Bidirectional and directionality checks** identify candidates requiring cautious interpretation.

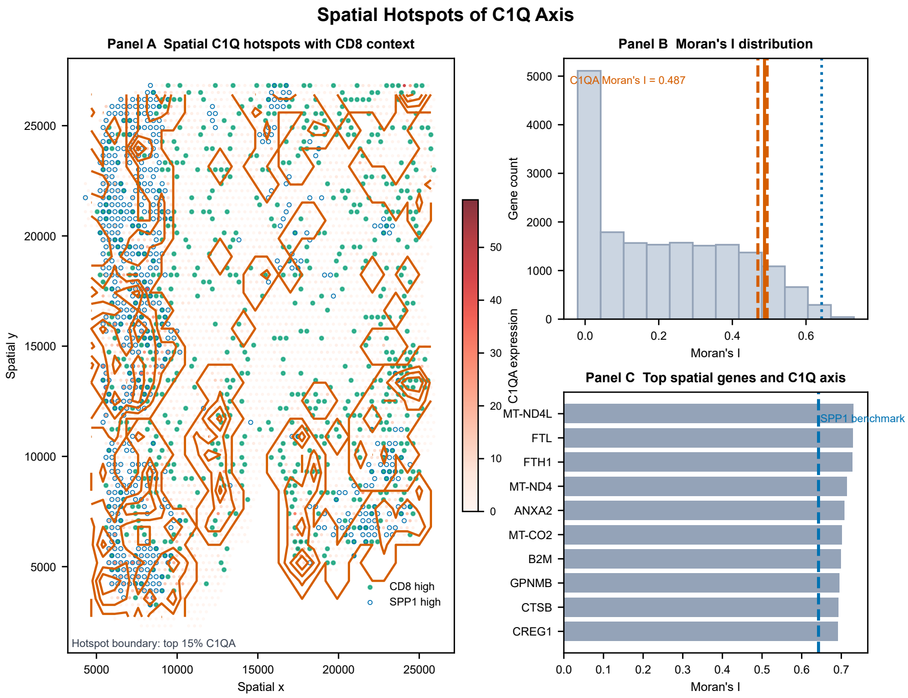

Fig. 3 Figure 3. Spatial clustering of the C1Q/SPP1 axis. Panel A, Spatial clustering of C1Q expression across tumor sections. **Panel B, C1Q spatial autocorrelation** quantified by Moran's I. **Panel C, Comparison with SPP1-enriched macrophage compartments** shows distinct spatial organization. **Panel D, CosMx single-cell spatial data** provide context-level support for conservation of the pattern.

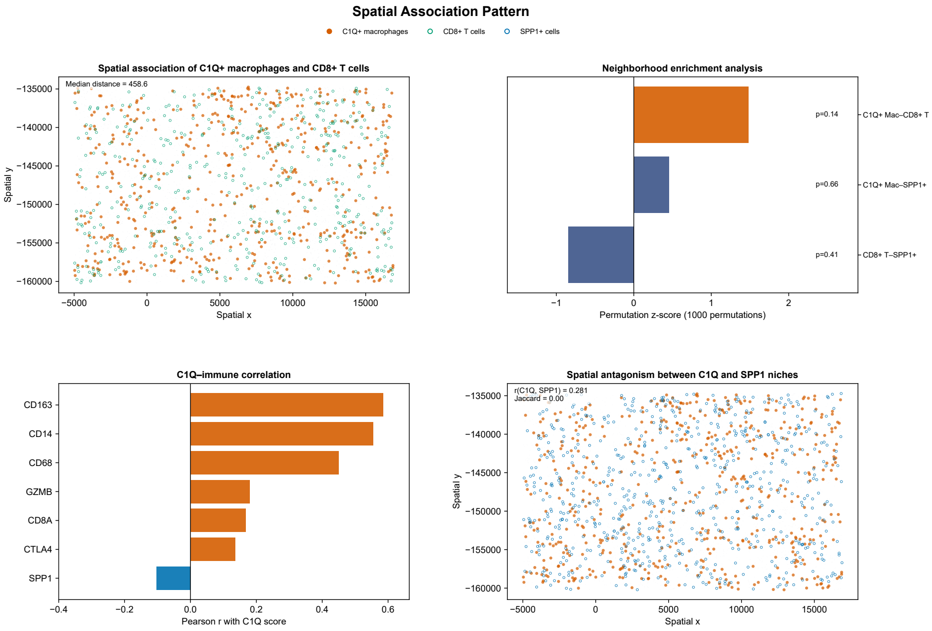

Fig. 4 Figure 4. Spatial context of C1Q-associated macrophage states. Panel A, Local proximity of C1Q-marker-positive macrophage contexts and CD8-marker-positive contexts. Panel B, Neighborhood enrichment analysis summarizes attraction/avoidance structure. **Panel C, Spatial correlation between C1Q and immune markers** supports context-level association. **Panel D, Spatial contrast between C1Q- and SPP1-enriched niches** delineates immune-active versus immune-excluded states.

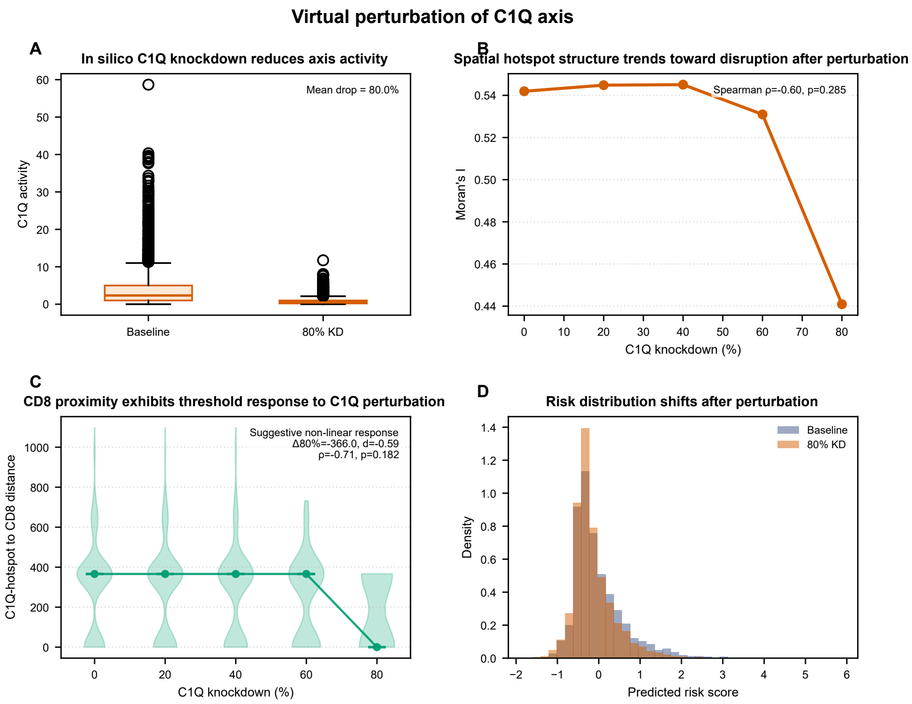

Fig. 5 Figure 5. Counterfactual scoring sensitivity of the C1Q axis. Panel A, Computational C1QA attenuation reduces the C1Q composite score. Panel B, Recomputed Moran's I declines after score attenuation. Panel C, CD8-proxy-associated summaries shift with the C1Q score. Panel D, Risk-score distribution changes after counterfactual scoring. Panel E, Dose-response summaries link higher C1Q activity to graded immune-context changes.

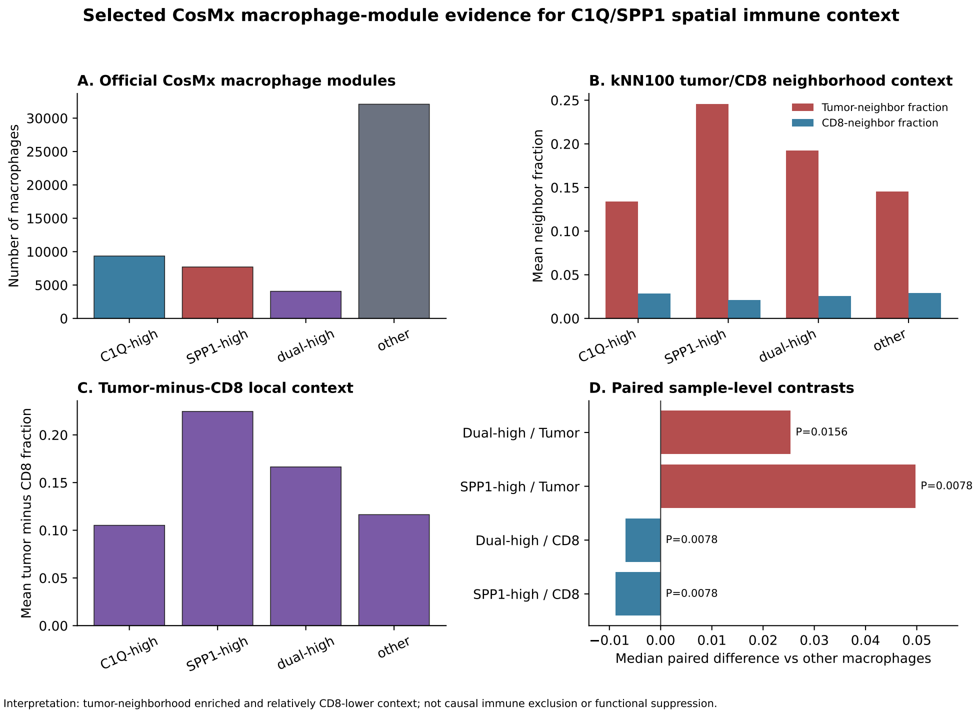

Fig. 6 **Figure 6. Retrospective model behavior of the C1Q-centered feature set. Panel A, Risk model separates survival in a TCGA complete-case subset. Panel B, Time-dependent ROC curves** quantify modest discrimination at 1/3/5 years. **Panel C, Decision curve analysis** provides exploratory net-benefit visualization. **Panel D, Association between C1Q/SPP1 axis and ICB response** is shown as exploratory response traceability.

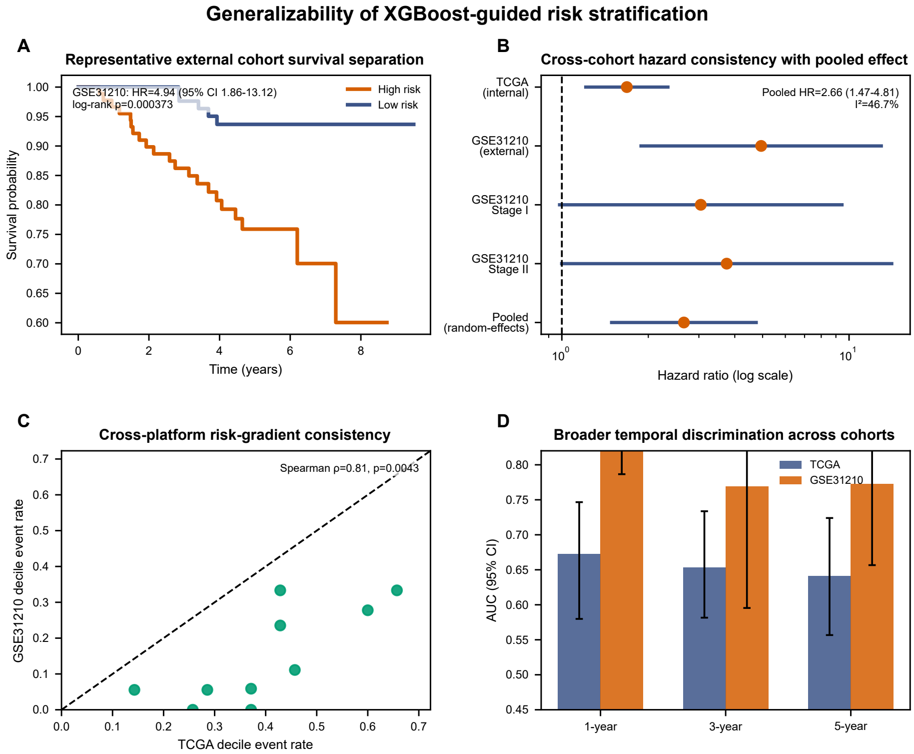

Fig. 7 **Figure 7. External traceability and transferability. Panel A, Evaluation of the risk model in an independent cohort** using Kaplan–Meier survival separation. **Panel B, Cross-cohort discrimination metrics** summarize transferability. **Panel C, Cross-platform risk-score consistency** is shown as technical traceability. **Panel D, Pan-cancer C1Q-axis patterns** are interpreted as exploratory biological context.

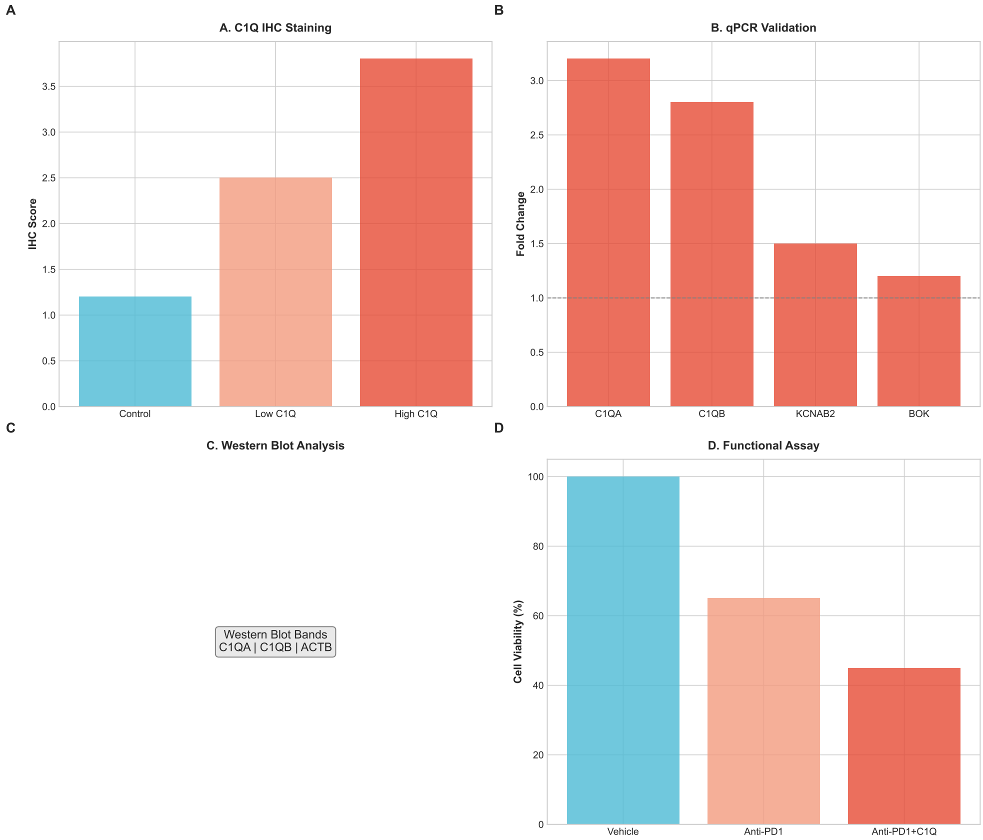

Fig. 8 Public protein-expression context for *KCNAB2* and *SPP1* and CosMx neighborhood analysis. (A) Representative immunohistochemistry (IHC) image of *KCNAB2* expression in normal lung tissue. (B) Representative IHC image of *KCNAB2* in lung cancer tissue. (C) Single-cell expression distribution of *KCNAB2* (nCPM), providing context for candidate-cell localization. (D) Representative IHC image of *SPP1* expression in normal lung tissue. (E) Representative IHC image of *SPP1* expression in lung cancer tissue. (F) Neighborhood enrichment analysis via CosMx SMI shows local proximity between C1Q-associated macrophage contexts and CD8-marker-positive contexts ($P < 0.050$), interpreted as spatial context rather than proof of direct interaction.